

YC APPLICATION · 2026

AI GUIDANCE FOR PHYSICAL WORK

ForgeAI

Expert walkthrough videos become real-time voice-guided coaching for any worker.



A senior technician records themselves doing a procedure once. ForgeAI watches the video, extracts step-by-step instructions, safety warnings, and Q&A — then coaches any worker through it in real time with voice guidance. Phone today, smart glasses tomorrow.

THE OPPORTUNITY

AI can't act in the world yet. But it can guide the human who does.

For physical work — field services, manufacturing, healthcare — AI can now see, reason, and coach in real time. This changes everything.

"Imagine wearing a small camera while an AI sees what you see and talks you through the job: 'turn off that valve', 'use the 3/8 inch wrench', 'that part looks worn, replace it'. Instead of needing months of training, workers become effective immediately."

— David Lieb, YC Partner

1**Multimodal AI works**

Gemini 2.0 Flash watches a 45-minute expert video and extracts structured procedures with exact specs — torque values, temperatures, part numbers. This wasn't possible 18 months ago.

2**Hardware is everywhere**

Every worker has a phone and earbuds. Our PWA installs in one tap. When smart glasses arrive, we're the software layer that powers them.

3**Labor crisis is urgent**

2.1M manufacturing jobs unfilled by 2030. Average electrician age: 55. Companies can't transfer decades of knowledge fast enough with traditional training.

2.1M

manufacturing jobs unfilled by
2030
(Deloitte)

\$47B

lost annually to inefficient
knowledge transfer

23%

of work time spent searching for
information
(IDC)

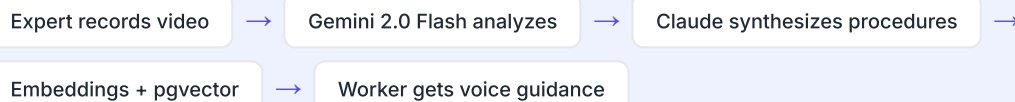
WHAT WE'VE BUILT

From expert video to voice-guided coaching — fully shipped.

Not a prototype. Not a demo. Every feature below is live and working today.

Expert Capture → Structured Knowledge → Real-Time Guidance

A senior technician records a walkthrough video once. Our AI pipeline watches the video, synthesizes it into structured procedures, and makes it available as voice-guided coaching — all within minutes.



Expert Video Capture

Upload a walkthrough video (MP4, WebM, MOV — up to 500MB). AI extracts procedures, equipment lists, safety warnings, and 8-20 Q&A pairs automatically.



Voice-Guided Procedures

Workers hear each step read aloud with safety warnings. Hands-free voice commands: "next", "repeat", "pause". No touching the phone.



Contextual Voice Q&A

Worker asks "what if the voltage reads 11?" mid-step. Gets an answer grounded in the expert's knowledge, read aloud. Full conversational loop.



Camera + RAG Chat

Point phone camera at a machine or part, ask a question. GPT-4o Vision + RAG searches your org's documents. Source-attributed answers.

TECH STACK

Next.js 16

React 19

Gemini 2.0 Flash

Claude Sonnet 4.5

GPT-4o

Whisper

pgvector

Prisma 7

Serwist PWA

EXPERT CAPTURE PIPELINE

One video in, structured knowledge out.

The expert capture pipeline turns a raw walkthrough video into training-quality documentation in 3-8 minutes — work that previously took weeks of technical writing.

1

Video Upload

Expert records a walkthrough showing the procedure — any camera, any format (MP4, WebM, MOV, MPEG). Up to 500MB. Uploaded to Vercel Blob.

2

Visual + Audio Analysis

Gemini 2.0 Flash watches every frame and listens to the narration. Extracts scenes, procedures, equipment, safety hazards, and timestamps.

3

Knowledge Synthesis

Claude Sonnet 4.5 synthesizes the raw analysis into a clean ExpertKnowledgeDocument: step-by-step procedures with exact specs (torque values, temperatures, pressures, part numbers).

4

Semantic Indexing

Pre-chunked by procedure step, Q&A pair, and safety note. Embedded with text-embedding-3-small (1536-dim vectors). Stored in PostgreSQL + pgvector.

5

Instantly Available

Procedure appears in the Guide Library. Workers can start following it immediately with voice guidance, or search it via RAG chat.

What the AI extracts

- ✓ **Procedures** — numbered steps with titles, descriptions, safety levels, timestamps
- ✓ **Equipment** — tools, machines, parts with part numbers and specs
- ✓ **Safety notes** — PPE, hazards, procedures, environmental warnings
- ✓ **Q&A pairs** — 8-20 realistic questions covering how-to, safety, troubleshooting, common mistakes
- ✓ **Full transcript** — complete audio transcription for searchability
- ✓ **Executive summary** — 2-3 paragraph overview of the procedure

Safety-aware by design

- ✓ **CRITICAL** — red badge, TTS prefixes "Warning. Critical safety step."
- ✓ **WARNING** — yellow badge, TTS prefixes "Caution."
- ✓ Auto-escalated for electrical, hydraulic, pneumatic, rotating, and chemical hazards
- ✓ Warnings displayed in highlighted callout boxes on each step

STEP-BY-STEP GUIDANCE

The worker experience: fully hands-free.

Worker opens a guide on their phone. Each step is read aloud. They say "next" when done. If confused, they ask a question and hear the answer. No hands needed.

1

Browse & start a guide

Worker opens the Guide Library, sees available procedures with step counts, safety badges, and expert names. Taps "Start Guide" or "Resume" to continue from where they left off.

2

Step-by-step with auto-play audio

Each step shows a progress bar, title, description, safety level, and warnings. The step description is automatically read aloud via TTS. Safety steps are prefixed with voice warnings.

3

Voice commands — no hands needed

Worker says "next" when done, "repeat" to hear it again, "pause" to stop audio, "go back" to revisit. Mic pauses during TTS to prevent feedback, then auto-resumes.

4

Ask anything mid-step

Any spoken phrase that isn't a command becomes a question. It goes to the contextual AI chat, gets a RAG-powered answer specific to this step, and the response is read aloud automatically.

5

Complete & track progress

On the last step, worker says "done" or taps "Complete". Session is logged. Managers can see procedure completion status. Workers can resume any procedure where they left off.

Voice commands

ACTION	SAY ANY OF THESE
Next step	next, continue, done, forward, skip
Previous	back, previous, go back
Read step	read, repeat, again, say again
Pause audio	pause, stop, mute, quiet
Resume	play, resume, start, speak
Ask question	Anything else → goes to AI chat

The hands-free loop

1. Step is read aloud
2. Worker performs the step
3. Says "next" → advances
- OR Asks a question → AI answers aloud
4. Mic auto-resumes after TTS
5. Repeat until complete

PLATFORM ROADMAP

Phone today. Smart glasses tomorrow. Same backend.

Our architecture is hardware-agnostic. Every expert video captured today becomes AR-ready content tomorrow without any rework.

✓ Phase 1 — Live**Phone + earbuds**

Worker launches procedure, listens to steps, uses voice commands. Camera capture for visual queries.

→ Phase 2 — Next**Design partners**

Onboard 2-3 field service companies. Refine voice UX with real workers on real jobs.

Phase 3**Smart glasses**

Same backend, same procedures — now overlaid on worker's field of view via Meta, XREAL, or Vuzix.

Phase 4**Proactive AI**

Continuous visual monitoring. AI detects what step the worker is on and warns about mistakes before they happen.

Our approach

- ✓ **Build the system, sell to companies** with existing workforces — field services, manufacturing, facilities
- ✓ **Vertical wedge:** start with companies already recording expert videos but storing them in SharePoint where nobody watches them
- ✓ **Hardware-agnostic:** phone is the wedge, glasses are the endgame — same content pipeline serves both
- ✓ **Compound moat:** every expert video captured makes the knowledge base more valuable over time

Market sizing

- ✓ **12.8M** manufacturing workers in the US alone
- ✓ **\$5.2B** field services market by 2028
- ✓ **\$28.4B** connected worker platform market by 2032 (Fortune Business Insights)
- ✓ Even **0.1%** of frontline workers at \$50/worker/month = **\$77M ARR**
- ✓ Large enterprise incumbents (PTC Vuforia, Honeywell Forge) cost \$500K+ — **SMBs are massively underserved**

\$25-50

per worker / month

Usage

per video minute processed for expert capture

Free

tier: 3 workers, 5 docs, 2 expert captures

SHIPPED FEATURES

Everything is live. Not a mockup.

We've built and shipped the complete pipeline — from video upload to voice-guided procedures with contextual AI assistance.



Expert Video Capture

Gemini 2.0 Flash + Claude Sonnet 4.5 pipeline. Extracts procedures, safety, equipment, Q&A.



Guide Library

Searchable grid of available procedures. Step count, safety badges, resume capability.



Voice-Guided Steps

Auto-play TTS with safety prefixes. Toggle on/off. Persisted preference.



Voice Commands

20+ command synonyms. Two-tier matching (short phrases vs. multi-word). Toast feedback.



Contextual Q&A

Step-aware RAG chat. Voice questions, streamed responses, read aloud automatically.



Camera + Vision

Point phone at machine, ask question. GPT-4o Vision analyzes with RAG context.



Document RAG

Upload PDFs, Word docs, images. Semantic search with source attribution on every answer.



PWA Mobile App

Installable on iOS & Android. Offline fallback. iOS-safe camera and audio capture.



Multi-Tenant Auth

Role-based access (admin/worker). Org-scoped data isolation. Email invite system.

What's built & what's next

Done

Expert Capture + Knowledge Base

Video → structured procedures → RAG-searchable knowledge

Done

Voice-Guided Procedures

Step-by-step audio guidance with hands-free commands + Q&A

Done

Mobile-First PWA

Camera, voice, offline fallback — works on any phone

Next

Design Partners

2-3 field service companies testing with real workers

Future

Smart Glasses Integration

AR overlay with same backend — Meta, XREAL, Vuzix

Session persistence

- ✓ Workers can **resume any procedure** exactly where they left off
- ✓ GuidanceSession tracks current step, total steps, status (in-progress / completed / abandoned)
- ✓ Auto-play preference persisted in localStorage
- ✓ Chat history per step — switch steps and conversation context follows
- ✓ Completion status tracked for reporting

YC APPLICATION · AI GUIDANCE FOR PHYSICAL WORK

The expert records it once. Every worker **knows it forever.**

ForgeAI turns expert walkthrough videos into real-time, voice-guided coaching. Phone today, smart glasses tomorrow. Fully built, fully shipped.



Expert captures

One video becomes structured procedures, safety notes, and Q&A — automatically



Hands-free guidance

Voice commands + contextual Q&A — workers never touch their phone



Glasses-ready

Same backend, same content — ready for AR overlay when glasses arrive

[Live demo:](#) forgeai-chi.vercel.app